

MOIKIPAA, Finland

WMO: 029210

Lat: 62.888N

Lon: 21.091E

Elev: 43

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-6.3	-17.1	2.2	-12.2	-11.0	3.0	-6.2	42.3	21.5	35.8	27.7	4.1	40	0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	69.4	63.1	66.9	61.3	64.8	59.9	64.1	67.7	62.4	66.0	60.8	64.0	9.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	84.4	66.3	60.6	79.2	64.6	59.1	75.0	63.0	29.3	67.8	28.1	66.2	27.0	64.2	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	32.6	29.3	26.3	DB	-11.4	74.3	7.6	4.4	-16.8	77.4	-21.2	80.0	-25.5	82.5	-31.0	85.7
(5)				WB	-11.7	66.5	7.4	2.9	-17.1	68.6	-21.4	70.4	-25.6	72.0	-31.0	74.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	18.2	19.8	25.8	33.8	44.7	54.2	59.9	57.7	50.5	43.0	33.4	24.7
	DBStd	16.17	13.73	11.65	8.11	6.04	5.84	5.30	3.77	4.21	4.25	4.64	7.75	9.29
	HDD50	4817	985	846	751	488	182	18	0	1	44	220	498	784
	HDD65	9530	1450	1266	1216	937	628	326	166	227	434	683	948	1249
	CDD50	772	0	0	0	0	19	143	307	241	60	2	0	0
	CDD65	12	0	0	0	0	0	2	8	2	0	0	0	0
	CDH74	17	0	0	0	0	0	2	14	1	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	13.2	13.7	13.4	13.3	12.6	11.9	11.3	10.8	11.8	13.4	15.7	16.1	14.5
Precipitation	PrecAvg	21.9	1.8	1.3	1.3	0.9	1.4	2.0	2.6	2.1	2.0	2.4	2.2	2.1
	PrecMax	27.6	3.0	2.6	2.1	1.6	3.0	4.1	4.2	4.0	4.3	4.4	5.1	3.9
	PrecMin	17.2	0.4	0.0	0.4	0.1	0.2	0.5	0.4	0.4	0.7	0.7	0.4	0.7
	PrecStd	3.1	0.8	0.7	0.5	0.4	0.9	0.9	1.2	1.2	1.1	1.0	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	37.7	38.1	54.7	65.6	71.2	75.5	70.6	63.2	54.2	46.5	39.6
		MCWB	35.6	35.9	35.9	48.1	58.8	63.1	68.7	63.8	57.1	51.7	44.7	37.7
	2%	DB	36.0	36.1	36.9	48.4	59.7	67.2	70.3	67.5	59.5	51.9	44.8	38.4
		MCWB	34.5	34.9	34.9	44.0	54.9	60.8	64.6	62.0	56.2	49.6	43.0	36.9
	5%	DB	34.7	34.9	35.7	44.0	55.7	64.4	67.8	65.4	57.7	50.4	43.6	37.2
		MCWB	33.5	34.0	34.2	41.0	50.5	58.5	62.1	61.1	54.9	48.7	41.8	36.0
	10%	DB	33.5	33.6	34.3	41.3	52.6	62.1	65.6	63.5	56.1	49.1	42.0	35.9
		MCWB	32.5	32.7	33.2	39.1	48.2	56.9	60.5	59.9	53.5	47.5	40.4	34.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	36.5	36.5	48.9	59.7	63.9	70.3	65.5	58.6	52.0	45.2	38.4
		MCDB	37.2	37.1	37.4	53.1	65.7	69.2	74.3	68.6	61.8	53.2	45.7	39.1
	2%	WB	34.8	35.3	35.4	44.0	55.6	61.8	65.4	63.7	56.8	50.3	43.5	37.1
		MCDB	35.5	35.9	36.3	47.6	59.7	66.4	70.0	66.3	58.7	51.5	44.6	37.8
	5%	WB	33.7	34.0	34.5	41.6	51.0	59.3	62.8	61.9	55.6	49.0	42.2	36.2
		MCDB	34.4	34.6	35.3	43.6	54.5	63.4	66.7	64.6	57.3	50.1	43.4	37.1
	10%	WB	32.6	32.8	33.3	39.3	48.7	57.3	61.4	60.2	54.2	47.8	40.8	34.9
		MCDB	33.4	33.5	34.2	41.3	51.8	61.0	64.8	62.9	55.8	49.0	41.8	35.9
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.6	8.6	7.9	8.0	7.3	7.1	6.3	5.6	4.9	5.4	8.1
		MCDBR	6.0	5.1	6.4	11.0	11.7	10.2	10.4	9.0	7.5	5.2	4.5	4.9
		MCWBR	6.1	5.1	5.3	7.6	6.9	5.8	6.2	6.1	5.7	4.5	4.5	4.7
	5% WB	MCDBR	5.9	4.7	5.1	10.6	9.4	9.9	9.1	8.4	7.1	5.0	4.3	4.8
		MCWBR	6.1	4.7	4.6	7.8	6.4	6.0	6.3	6.2	6.0	4.5	4.5	4.9
Clear-Sky Solar Irradiance	taub	0.244	0.298	0.318	0.358	0.356	0.352	0.374	0.362	0.336	0.319	0.387	0.350	
	taud	2.079	2.201	2.281	2.324	2.439	2.485	2.448	2.472	2.524	2.387	2.686	2.550	
	Ebn at Noon	125	202	248	261	273	276	266	259	245	196	115	70	
	Edh at Noon	10	20	27	32	31	30	31	28	22	17	9	6	
All-Sky Solar Radiation	RadAvg	47	225	703	1213	1696	1899	1733	1327	782	301	73	25	
	RadStd	6	44	83	125	158	131	147	113	72	58	14	2	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page